

AMOGH RAMAGIRI

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EDUCATION

George Washington University

Master of Science in Data Science (CCAS Dean's Award)

Aug 2024 – May 2026

Washington, DC

PROFESSIONAL EXPERIENCE

Fulton Bank

May 2025 – Dec 2025

Data Scientist

Lancaster, PA

- Built **mortgage default prediction** model (Random Forest) on **1M+ loan records**, achieving **91% recall** on high-risk applications and reducing potential portfolio losses by **18%** (~\$2M+ annually).
- Optimized **20+ SQL queries** for credit risk reporting, cutting **execution time 15%** and eliminating 4+ hours of weekly manual data reconciliation across the Lending team.
- Led **Azure cloud migration** (Data Factory, Synapse Analytics), improving data quality **20%** and accelerating pipeline throughput **43%**, enabling real-time reporting.
- Redesigned **37+ Power BI dashboards** on Microsoft Fabric, consolidating 6+ data sources and cutting **executive reporting cycle from weekly to daily**.

Factocart, Velabh Technologies Pvt. Ltd.

Mar 2024 – Aug 2024

Data Scientist

Bengaluru, India

- Processed **2M+ transaction records** from AWS RDS using **PySpark** and **SQL**; built **demand forecasting models** (Scikit-learn) achieving **92% accuracy**, reducing stockouts by **13%** across 10+ product categories.
- Designed and executed **A/B experiments** and hypothesis tests across marketing campaigns, surfacing strategies that drove **15% lift in conversion rates** and **11% revenue growth** across 3 business units.
- Built **LLM-powered RAG pipeline** (LangChain, OpenAI API, Pinecone), cutting analyst review time **60%** and reducing API costs **35%** via prompt chaining and few shot optimization.

Embedded System Lab, NCUE

Sep 2023 – Mar 2024

Research Scientist

Changhua, Taiwan

- Engineered **multi-modal biometric authentication** system (DeepFace CNN + MFCC audio features, PyTorch) for offline edge deployment on Raspberry Pi, improving authentication accuracy **30%** over single-modal baseline.
- Reduced **false rejection rate 25%** via statistical root-cause analysis on **10,000+ authentication failures**, identifying and resolving 4 critical failure modes across lighting and audio variance conditions.
- Architected **edge-to-cloud data pipeline** (Flask, REST APIs) streaming biometric records from Raspberry Pi cluster to centralized training infrastructure with **99% data integrity** and sub-200ms latency.

Wodo

Jun 2022 – Sep 2023

Data Analyst

Bengaluru, India

- Automated **marketing performance reporting pipeline** in Python and SQL across **13+ client accounts**, consolidating data from 5+ ad platforms into a unified dashboard and reducing weekly reporting time by **~75%**.
- Built **real-time engagement and purchase dashboards** enabling clients to reallocate marketing budgets dynamically, reducing **wasted ad spend by 20%** and improving ROAS across paid channels.
- Conducted **customer segmentation** using K-Means and hierarchical clustering on behavioral data, identifying 5 high-value audience cohorts that improved campaign targeting precision and **lifted conversion rates 12%**.

TECHNICAL SKILLS

Programming and APIs: Python, R, SQL, FastAPI, Flask, RESTful APIs

AI/ML Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, SHAP

GenAI and NLP: LangChain, OpenAI API, Pinecone, RAG, BERT, Sentence Transformers, spaCy, NER, Text Classification, TF-IDF, Embeddings

Cloud and MLOps: AWS (EC2, S3, RDS, SageMaker, EMR), Azure (Data Factory, Synapse), Docker, MLflow, CI/CD, Linux, Git

Data and Visualization: PostgreSQL, MongoDB, SSMS, PySpark, Pandas, NumPy, Tableau, Power BI

PROJECTS

Ticket Triage Knowledge Base Gap Analysis and Generation

2026

- Deployed production **agentic NLP system** serving **44,000+ users**, autonomously processing **100,000+ IT support tickets** via classification, semantic clustering, and vector search, identifying **120+ recurring issue patterns**.
- Integrated **Azure OpenAI GenAI** summarization layer with **RAG pipeline**, reducing IT triage analyst effort by **43%** and enabling automatic knowledge base article generation for top recurring issues.

Dynamic Pricing Optimization using Machine Learning

2025

- Engineered **price elasticity and demand features** from **500,000+ transaction records**; deployed **XGBoost** price sensitivity model achieving **10.7% revenue lift** through data driven pricing recommendations.
- Improved **demand forecasting accuracy by 18%** over baseline using gradient boosting with custom behavioral features, enabling real time **price optimization** across product tiers.